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Research Proposal

*Image-Based Restricted Dog Breed Classification Using CNN Transfer Learning
and the Stanford Dogs Dataset*

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Contents

1	Introduction: Why This Matters	6
1.1	Scope and Limitations of the Project	7
1.2	The Broader Context: AI in Classification	8
2	Research Objectives and Hypotheses	9
2.1	Primary Objective	9
2.2	Secondary Objectives	9
2.3	Hypotheses	10
2.4	Alignment with the Real World	11
3	Literature Review	12
3.1	The ImageNet Story: From Feature Engineering to Deep Learning	12
3.2	Transfer Learning: Standing on the Shoulders of ImageNet	13
3.3	Dog Breed Classification: The State of the Art	14
3.4	Interpretability: Grad-CAM and the Trust Problem	14
3.5	Algorithmic Fairness and Bias	15
4	Sampling Strategy	16
4.1	Source Dataset	16
4.2	Label Engineering: From 120 Breeds to Two Classes	16
4.3	Stratified Splitting	17
4.4	Dataset Size and the Scaling Question	17
4.5	Ethical Sourcing	17
5	Methodology	18
5.1	Notebook 0: Data Preparation	19
5.2	Notebook 1: The Architecture Timeline	19

5.3	Notebook 2: Modern Transfer Learning Zoo	19
5.4	Notebook 3: Hyperparameter Search	20
5.5	Notebook 4: Evaluation and Threshold Optimisation	20
5.6	Notebook 5: Interpretability and Grad-CAM	21
5.7	Regularisation Techniques: What We Search For	21
5.8	Why Not Vision Transformers?	22
6	Ethical and Data Protection Considerations	23
6.1	The Stakes of Misclassification	23
6.2	Data Sourcing and Compliance	23
6.3	Bias Awareness	24
6.4	Transparency and Interpretability	24
6.5	Deployment Safeguards	24
7	Expected Challenges and Mitigation Strategies	25
7.1	Challenge 1: Visual Similarity Between Classes	25
7.2	Challenge 2: Heterogeneous Positive Class	25
7.3	Challenge 3: Limited Dataset Size	26
7.4	Challenge 4: Computational Resources	26
7.5	Challenge 5: Missing Breeds	26
8	Conclusion	27
8.1	Timeline	28

List of Figures

1	The American XL Bully — banned outright in Ireland since October 2024. Ownership requires a Certificate of Exemption from February 2025.	6
2	The English Bull Terrier — one of eleven restricted breeds under Irish law. Its distinctive egg-shaped head makes it relatively identifiable, but crossbred dogs present far greater challenges.	9
3	The Japanese Akita — restricted under Irish law. Akitas are a large, powerful breed originally bred for guarding. Their thick double coat and curled tail are distinctive, but they share body mass and posture with several unrestricted breeds.	11
4	The Rottweiler — one of the more visually distinctive restricted breeds. Preliminary experiments showed it was the easiest breed for CNNs to classify, consistent with its strong morphological signal (black-and-tan colouring, blocky head, muscular build).	14
5	The Rhodesian Ridgeback — a restricted breed known for its distinctive dorsal ridge of backward-growing hair. In preliminary experiments, it achieved the highest per-breed classification accuracy (80.6%), likely due to this unique visual feature. .	16
6	The Japanese Tosa — a restricted breed not present in the Stanford Dogs Dataset. Its absence illustrates a key limitation: the model can only classify breeds it has seen. Future work could incorporate web-scraped images of under-represented breeds.	18
7	The Doberman Pinscher — restricted under Irish law. In breed-specific binary classifiers, the Doberman achieved a validation accuracy of 77.7%, placing it in the middle of the difficulty range.	20
8	The Staffordshire Bull Terrier — restricted under Irish law and frequently confused with the American Pit Bull Terrier and the American Staffordshire Terrier. This morphological ambiguity is the central ethical challenge: a misclassification can have life-or-death consequences for the animal.	23
9	Two restricted breeds with very different morphology. The model must learn that “restricted” is a legal category, not a visual one — these dogs share a legal status but not a body plan.	25
10	The Bandog — a restricted breed that is not a single standardised breed but a type of large, muscular cross, historically bred as a guard dog. The absence of a standardised phenotype makes visual classification particularly challenging, even for human experts.	27

1 Introduction: Why This Matters

On the afternoon of 4 June 2024, a child was attacked by an American XL Bully in Wexford. Within weeks, the Irish Government announced emergency legislation. By October 2024, importing, breeding, selling, or rehoming XL Bully dogs became illegal. By February 2025, ownership itself became prohibited without an exemption certificate requiring microchipping, licensing, and neutering (RTÉ News, 2024).



Figure 1: The American XL Bully — banned outright in Ireland since October 2024. Ownership requires a Certificate of Exemption from February 2025.

But there is a deeper problem. Ireland’s breed-specific legislation does not only cover the XL Bully. The Control of Dogs Regulations 1998 (Government of Ireland, 1998) designates eleven additional breeds as *restricted*: dogs that are not banned outright but must be muzzled, leashed (no longer than two metres), and handled by a person over sixteen years of age in all public places. The restricted breeds are:

- American Pit Bull Terrier
- Bull Mastiff
- Doberman Pinscher
- English Bull Terrier
- German Shepherd (Alsatian)
- Japanese Akita

- Japanese Tosa
- Rhodesian Ridgeback
- Rottweiler
- Staffordshire Bull Terrier
- Bandog

Enforcement of these regulations depends almost entirely on a dog warden or Garda officer looking at a dog and making a visual judgement. That judgement is subjective, inconsistent, and prone to error — particularly with crossbred dogs, dogs in poor condition, or breeds that share similar morphology (Clarke, 2018; Voith et al., 2009). A Staffordshire Bull Terrier and an American Pit Bull Terrier can look nearly identical to an untrained eye. A Labrador–Rottweiler cross may or may not trigger enforcement depending on which parent’s features are more prominent.

The consequences of misidentification are severe. A wrongly classified dog can be seized, quarantined, or destroyed. An owner who does not muzzle a restricted breed faces prosecution. The human cost is real: families separated from pets, dogs euthanised on the basis of appearance rather than behaviour.

This project asks a simple question: **can a computer do better?**

Specifically, it investigates whether convolutional neural networks (CNNs) — the deep learning architectures that have dominated image classification since 2012 — can learn to distinguish restricted from unrestricted dog breeds using photographs alone. The answer matters not just technically but ethically. If an algorithm can achieve higher accuracy and greater consistency than a human assessor, it could serve as a decision-support tool: not replacing the warden, but giving them a second opinion grounded in data rather than intuition.

1.1 Scope and Limitations of the Project

It is important to be clear about the scope of this study. This project does not claim that breed classification is, by itself, a valid basis for dangerous-dog policy; that is a separate ethical, veterinary and legal debate, well documented in the literature (Clarke, 2018). Nor does the project propose a fully autonomous decision-making system. Instead, it offers a focused technical investigation: given the current legal reality of breed-specific legislation, can machine learning support more consistent and evidence-informed classification of restricted dog breeds?

1.2 The Broader Context: AI in Classification

Image classification powered by deep learning has transformed fields from medical imaging (detecting tumours in X-rays) to agriculture (identifying crop diseases from photographs) to everyday life (PlantNet, the app that identifies plants from a phone camera) (PlantNet, 2025). The core technology — transfer learning from models pre-trained on millions of images — is well understood and accessible. What varies is the quality of the question being asked, the rigour of the data preparation, and the honesty of the evaluation.

This project sits at that intersection: a well-defined societal question (can we reliably classify restricted breeds?), a publicly available dataset (the Stanford Dogs collection of 20,000 labelled images), and a transparent methodology (systematic comparison of CNN architectures with interpretable outputs).

2 Research Objectives and Hypotheses



Figure 2: The English Bull Terrier — one of eleven restricted breeds under Irish law. Its distinctive egg-shaped head makes it relatively identifiable, but crossbred dogs present far greater challenges.

2.1 Primary Objective

To evaluate and compare the performance of multiple CNN architectures — spanning the full history of the ImageNet competition from AlexNet (2012) to modern architectures like Inception-ResNetV2 and NASNetMobile — on the binary classification of restricted versus unrestricted dog breeds under Irish legislation.

This is not a single-model study. It is a controlled experiment across six to ten architectures, using identical data, identical evaluation metrics, and identical training conditions, to answer a specific empirical question: *does architectural progress matter for this task, or does the size and quality of the dataset matter more?*

2.2 Secondary Objectives

1. **Curate a clean, reproducible dataset** from the Stanford Dogs collection, with legally grounded label engineering mapping each breed to its Irish regulatory status.
2. **Systematically search for optimal hyperparameters** rather than hand-tuning. Dropout

rates, learning rates, augmentation intensity, and fine-tuning depth will be explored using Keras Tuner’s Hyperband algorithm across 50–100 automated trials.

3. **Evaluate using metrics appropriate for imbalanced classification.** Accuracy alone is misleading when classes are unequal. This project will report precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and PR-AUC, alongside threshold optimisation to calibrate the trade-off between false positives and false negatives.
4. **Interpret model decisions using Grad-CAM** (Selvaraju et al., 2017) to visualise which regions of a dog image the CNN focuses on when making its prediction. This is essential for trust: a model that looks at ears and snouts is plausible; a model that looks at grass or fencing is not.
5. **Tell the story of deep learning itself.** By running models from 2012 (AlexNet) through 2017 (SENet) and beyond, the project serves as a pedagogical timeline of how CNNs evolved — from 8-layer networks with 60 million parameters to 152-layer residual networks that surpassed human performance on ImageNet.

2.3 Hypotheses

H1: Pre-trained CNN architectures using transfer learning will significantly outperform CNNs trained from scratch on the same restricted-breed dataset, achieving validation accuracy above 85% compared to the 62–67% baseline observed in preliminary experiments.

H2: Automated hyperparameter search will identify configurations that outperform hand-tuned defaults, with the strongest improvements in recall for the restricted class (the safety-critical metric — missing a restricted dog is worse than falsely flagging an unrestricted one).

H3: Grad-CAM visualisations will confirm that high-performing models attend to morphologically meaningful features (head shape, ear position, body proportion) rather than dataset artefacts (background, image quality, photographer angle).

2.4 Alignment with the Real World



Figure 3: The Japanese Akita — restricted under Irish law. Akitas are a large, powerful breed originally bred for guarding. Their thick double coat and curled tail are distinctive, but they share body mass and posture with several unrestricted breeds.

These objectives are deliberately practical. A dog warden in Thurles or Tralee does not need a Vision Transformer trained on 300 million parameters. They need a tool that works on a phone, gives a confidence score, and shows them *why* it thinks a dog is restricted. Every design decision in this project — from the choice of lightweight architectures to the emphasis on interpretability — is guided by that end use.

3 Literature Review

This review traces three interconnected threads: the evolution of CNN architectures through the ImageNet competition, the application of transfer learning to small-dataset problems, and the specific challenges of dog breed classification in legal contexts.

3.1 The ImageNet Story: From Feature Engineering to Deep Learning

The ImageNet Large Scale Visual Recognition Challenge (ILSVRC), run annually from 2010 to 2017, is the single most important benchmark in the history of computer vision. Its trajectory tells the story of deep learning’s rise.

Table 1: ILSVRC competition winners, 2010–2017. Human top-5 error rate is approximately 5.1%.

Year	Winner	Era	Top-5 Error	Depth	Key Innovation
2010	NEC-UIUC	Feature eng.	28.2%	—	SIFT + Fisher vectors + SVM
2011	XRCE	Feature eng.	25.8%	—	Compressed Fisher vectors
2012	AlexNet	Deep learning	15.3%	8	ReLU, dropout, multi-GPU
2013	ZFNet	Deep learning	11.2%	8	Smaller filters (7×7), feature visualisation
2014	GoogLeNet	Deep learning	6.7%	22	Inception module — parallel filter paths
2014	VGGNet*	Deep learning	7.3%	16–19	Deep stacks of 3×3 convolutions
2015	ResNet	Deep learning	3.6%	152	Skip connections; surpassed human performance
2016	Trimps-Soushen	Deep learning	3.0%	—	Ensemble; no single canonical architecture
2017	SENet	Deep learning	2.3%	152+	Channel attention — learns which features

*Runner-up in 2014 but equally influential as the winner.

Before 2012, the competition was won by classical machine learning methods: hand-crafted features (SIFT descriptors, histogram of oriented gradients) fed into support vector machines (Goodfellow et al., 2016). Error rates hovered around 25–28%. Then AlexNet arrived.

AlexNet (2012) was the shock that launched the deep learning revolution. Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton entered an eight-layer convolutional neural network that achieved a top-5 error of 15.3% — nearly ten percentage points ahead of the second-place entry (Krizhevsky et al., 2012). AlexNet introduced three ideas that remain foundational: the ReLU activation function (replacing the slower sigmoid), dropout regularisation (randomly zeroing neurons during training to prevent overfitting), and training across multiple GPUs.

ZFNet (2013) refined AlexNet with smaller convolutional filters (7×7 instead of 11×11), preserving more spatial information in early layers. Its more lasting contribution was a technique for

visualising what CNNs learn at each layer — demonstrating that early layers capture edges and textures, while deeper layers respond to complex object parts.

The 2014 double act is the most interesting year in the competition’s history. GoogLeNet (Inception V1) won with a 6.7% error rate by introducing the Inception module: instead of choosing a single filter size at each layer, it applies 1×1 , 3×3 , and 5×5 convolutions in parallel and concatenates the results. This produced a deeper network (22 layers) with far fewer parameters (6 million) than its competitors (Szegedy et al., 2015). VGGNet, the runner-up at 7.3%, took the opposite approach: extreme simplicity, stacking sixteen to nineteen layers of 3×3 convolutions. With 138 million parameters, VGG was computationally expensive but became the default backbone for transfer learning because its regular structure made it easy to adapt (Simonyan and Zisserman, 2014).

ResNet (2015) solved the problem that had limited all previous architectures: the vanishing gradient. As networks grow deeper, gradients shrink during backpropagation, making early layers nearly impossible to train. Kaiming He’s solution was the residual connection — a shortcut that adds the input of a block directly to its output, allowing gradients to flow through identity paths (He et al., 2016). This enabled networks of 50, 101, and even 152 layers, achieving a 3.57% top-5 error that surpassed human performance (estimated at 5.1%) for the first time.

Why the competition stopped. After 2017, the ILSVRC was discontinued. The problem had been largely solved — error rates below 2.3% left little room for architectural competition. Research shifted toward efficiency (MobileNet, EfficientNet), self-supervised learning, and Vision Transformers (Dosovitskiy et al., 2020). For a binary classification task like ours, the lesson is clear: the architectures available through standard libraries (VGG16, ResNet50, InceptionV3, Xception, InceptionResNetV2, NASNetMobile) represent decades of engineering and are more than sufficient.

3.2 Transfer Learning: Standing on the Shoulders of ImageNet

The reason these architectures are useful for dog breed classification is transfer learning. Rather than training a CNN from scratch on our ten thousand images (which would overfit quickly), we take a network that has already been trained on 1.2 million ImageNet images across 1,000 categories. The early layers — which learn universal features like edges, textures, and colour gradients — are frozen. Only the final classification layers are retrained on our specific task (Yosinski et al., 2014).

This approach is well supported by the literature. Yosinski et al. (2014) demonstrated that the first few layers of CNNs trained on ImageNet learn features that generalise across tasks, while later layers become increasingly task-specific. Tan and Le (2019) showed that transfer learning from EfficientNet achieves state-of-the-art results on fine-grained classification tasks with as few

as 5,000 training images.

The practical implication is that architectural depth and pre-trained weight quality matter more than dataset size for our task. A VGG16 or ResNet50, pre-trained on ImageNet, arrives at our problem already knowing what fur, eyes, ears, and body proportions look like. Our fine-tuning merely teaches it which combinations of those features correspond to “restricted” versus “unrestricted.”

3.3 Dog Breed Classification: The State of the Art

The Stanford Dogs Dataset (Khosla et al., 2011), containing 20,580 images across 120 breeds, is the standard benchmark for fine-grained dog breed classification. Liu et al. (2012) demonstrated that fine-tuning VGG16 on this dataset yields accuracy above 85%, while more recent work with EfficientNet variants has pushed accuracy above 90% on the full 120-class problem.

However, binary classification of *restricted versus unrestricted* breeds is a different problem. It requires the model to learn a legal boundary, not a biological one. A Rottweiler and a Doberman are both restricted but look very different. A Staffordshire Bull Terrier (restricted) and a Boxer (unrestricted) look very similar. This makes the task harder in some respects and easier in others — the model must learn a heterogeneous positive class defined by law rather than morphology.

No published study could be found that applies deep learning specifically to the classification of legally restricted breeds under Irish or UK legislation. This gap motivates the current project.

3.4 Interpretability: Grad-CAM and the Trust Problem



Figure 4: The Rottweiler — one of the more visually distinctive restricted breeds. Preliminary experiments showed it was the easiest breed for CNNs to classify, consistent with its strong morphological signal (black-and-tan colouring, blocky head, muscular build).

A model that achieves 90% accuracy but makes its decisions by looking at the grass behind the dog is not useful. Grad-CAM (Gradient-weighted Class Activation Mapping) (Selvaraju et al., 2017) addresses this by producing a heatmap showing which regions of an input image most influenced the prediction. For each image, three panels are generated: the original photograph, the Grad-CAM heatmap, and the heatmap overlaid on the original.

This is not just a technical nicety — it is an ethical requirement. In a regulatory context, any automated tool must be explainable. A dog warden shown a prediction alongside a heatmap highlighting the dog’s head and shoulders can make an informed judgement. A prediction without explanation is a black box that cannot be challenged.

3.5 Algorithmic Fairness and Bias

Buolamwini and Gebru (2018) demonstrated that commercial facial recognition systems performed significantly worse on darker-skinned faces — a bias inherited from unbalanced training data. The same logic applies to animal classification: if the training set contains more images of Rottweilers than Bull Mastiffs, the model will be better at recognising Rottweilers. If most images show dogs in gardens with green backgrounds, the model may associate grass with “restricted.”

This project addresses bias through stratified sampling (ensuring each breed is represented proportionally in train, validation, and test splits), data augmentation (random flips, rotations, and zoom to reduce background dependence), and post-hoc analysis (Grad-CAM verification that the model attends to the dog, not the setting).

4 Sampling Strategy



Figure 5: The Rhodesian Ridgeback — a restricted breed known for its distinctive dorsal ridge of backward-growing hair. In preliminary experiments, it achieved the highest per-breed classification accuracy (80.6%), likely due to this unique visual feature.

4.1 Source Dataset

The Stanford Dogs Dataset (Khosla et al., 2011) provides the foundation: 20,580 images across 120 breeds, sourced from ImageNet and annotated with breed-level labels. The dataset was selected for three reasons: it is publicly available under a research licence, it is widely used in the literature (enabling comparison with published results), and it contains sufficient examples of most Irish restricted breeds.

4.2 Label Engineering: From 120 Breeds to Two Classes

The core task is binary: *restricted* versus *unrestricted*. The restricted class is constructed by mapping the eleven breeds listed in the Control of Dogs Regulations 1998 (Government of Ireland, 1998) to their Stanford Dogs equivalents. Not all restricted breeds appear in the dataset — the American Pit Bull Terrier, Japanese Tosa, Bandog, and XL Bully are absent from Stanford Dogs. The breeds successfully mapped are:

- Bull Mastiff, Doberman Pinscher, German Shepherd, Rhodesian Ridgeback, Rottweiler, Staffordshire Bull Terrier

The unrestricted class is constructed by randomly sampling 24 breeds from the remaining 114, using a fixed random seed (58) for reproducibility. This deliberate over-representation of the unrestricted class (24 breeds versus 6) reflects the real-world prior: most dogs encountered by a warden will not be restricted.

4.3 Stratified Splitting

The combined dataset is split 60/20/20 into training, validation, and test sets using stratified random sampling at the breed level. This ensures that every breed appears in every split, preventing the model from encountering unseen breeds at evaluation time — a form of data leakage that would inflate apparent performance.

4.4 Dataset Size and the Scaling Question

A key control experiment addresses the question: *how much data is enough?* The full dataset will be compared against 75% and 50% subsets, with performance metrics logged at each level. If accuracy degrades gracefully, the model is robust; if it collapses, the dataset is at the minimum viable size. This informs any future deployment where additional labelled images may not be available.

4.5 Ethical Sourcing

All images are sourced from the Stanford Dogs Dataset, which is derived from ImageNet and available for academic research. No personally identifiable information is present in the dataset. No user-generated content is collected. The project complies with GDPR principles of data minimisation and purpose limitation (Voigt and von dem Bussche, 2017).

5 Methodology



Figure 6: The Japanese Tosa — a restricted breed not present in the Stanford Dogs Dataset. Its absence illustrates a key limitation: the model can only classify breeds it has seen. Future work could incorporate web-scraped images of under-represented breeds.

The methodology is structured as a six-notebook pipeline, each building on the previous one, designed to run on Google Colab with GPU acceleration and persistent storage via Google Drive.

5.1 Notebook 0: Data Preparation

The Stanford Dogs dataset is downloaded, extracted, and normalised. Breed folders are renamed from ImageNet synset notation (e.g., n02106662-German_Shepherd) to readable slugs (german_shepherd). Images are renamed sequentially within each breed. The restricted/unrestricted label mapping is applied, and the 60/20/20 stratified split is executed. A breed distribution analysis confirms balanced representation across splits.

5.2 Notebook 1: The Architecture Timeline

This notebook tells the story of ImageNet by training the canonical competition winners on our dataset: AlexNet (2012), VGG16 (2014), GoogLeNet/InceptionV3 (2014), and ResNet50 (2015). All use frozen ImageNet pre-trained weights with identical classification heads (GlobalAveragePooling2D, Dropout, Dense sigmoid). The purpose is pedagogical and comparative: how does 2012-era transfer learning compare with 2015-era transfer learning on the same binary task?

5.3 Notebook 2: Modern Transfer Learning Zoo

Six state-of-the-art architectures are trained under identical conditions:

Table 2: Model zoo: six architectures compared under identical conditions.

Architecture	Input Size	Parameters	Year
VGG16	224×224	138M	2014
ResNet50	224×224	25M	2015
InceptionV3	299×299	24M	2015
Xception	299×299	23M	2016
InceptionResNetV2	299×299	56M	2016
NASNetMobile	224×224	5M	2018

All models use frozen ImageNet backbones with a shared classification head: GlobalAveragePooling2D → Dropout(0.2) → Dense(1, sigmoid). Training runs for 22 epochs with early stopping (patience 3), Adam optimiser at 10^{-4} , and data augmentation (random horizontal flips, 5% rotation, 10% zoom). Every epoch is logged to CSV. Model checkpoints are saved on validation loss improvement.

5.4 Notebook 3: Hyperparameter Search

Rather than guessing optimal hyperparameters, this notebook uses Keras Tuner’s Hyperband algorithm to search over:

- **Dropout rate:** 0.2, 0.3, 0.4, 0.5
- **Learning rate:** 3×10^{-5} , 1×10^{-4} , 3×10^{-4}
- **Fine-tuning depth:** freeze all, unfreeze last 10%, 20%, 30% of base layers
- **Augmentation intensity:** light (current), moderate (15% rotation, 20% zoom, brightness jitter), strong (30% rotation, random erasing)

Hyperband is a principled early-stopping strategy for hyperparameter search: it allocates a small budget to many configurations, then progressively increases the budget for the most promising ones. Fifty to one hundred trials will be executed, with each trial lasting up to 8 epochs. The best configuration is then retrained for a full 30 epochs. All trial results are saved as CSV and JSON for the final report.

5.5 Notebook 4: Evaluation and Threshold Optimisation



Figure 7: The Doberman Pinscher — restricted under Irish law. In breed-specific binary classifiers, the Doberman achieved a validation accuracy of 77.7%, placing it in the middle of the difficulty range.

Standard evaluation reports accuracy, but for an imbalanced binary problem with real-world consequences, accuracy is insufficient. This notebook computes:

- **Precision:** of dogs predicted as restricted, how many actually are? (False positives cause unnecessary enforcement.)
- **Recall:** of dogs that are actually restricted, how many does the model catch? (False negatives are a safety risk.)
- **F1-score:** the harmonic mean of precision and recall.
- **Matthews Correlation Coefficient (MCC):** a balanced metric that accounts for all four cells of the confusion matrix, recommended for imbalanced datasets (Chicco and Jurman, 2020).
- **PR-AUC:** area under the precision–recall curve, more informative than ROC-AUC when classes are imbalanced.
- **Threshold optimisation:** the default decision threshold is 0.5, but this may not be optimal. By sweeping thresholds from 0.1 to 0.9 and plotting precision, recall, and F1 at each, we can identify the threshold that best balances safety (recall) with fairness (precision).

5.6 Notebook 5: Interpretability and Grad-CAM

For the best-performing model, Grad-CAM heatmaps are generated for one representative image per breed in the test set. Each output is a three-panel figure: original image, heatmap, and superimposed overlay. These visualisations are saved at 150 DPI for inclusion in the final report.

The notebook also performs a *failure case analysis*: examining the images where the model was most confident but wrong. This reveals systematic biases (e.g., does the model struggle with particular backgrounds, poses, or breeds?) and informs future data collection priorities.

5.7 Regularisation Techniques: What We Search For

The following regularisation and augmentation techniques are not assumed — they are searched over:

Table 3: Regularisation and augmentation techniques explored in the hyperparameter search.

Technique	Purpose
Dropout	Randomly zeros neurons during training; prevents co-adaptation of features
Batch normalisation	Standardises layer activations per mini-batch; stabilises and accelerates training
L2 weight decay	Penalises large weights; encourages simpler, more generalisable models
Early stopping	Halts training when validation loss stops improving; prevents overfitting
Random horizontal flip	Dogs face left or right equally; removes directional bias
Random rotation	Simulates tilted photographs; improves robustness to camera angle
Random zoom	Simulates distance variation; teaches the model to recognise dogs at different scales
Random brightness	Simulates lighting conditions; reduces sensitivity to exposure
Random erasing	Occludes random patches; forces the model to use multiple features, not just one

The key principle is that these are not applied blindly. Each technique has a cost (computational time, potential under-fitting if too aggressive) and a benefit (reduced overfitting, better generalisation). The hyperparameter search finds the optimal combination for this specific dataset.

5.8 Why Not Vision Transformers?

Vision Transformers (ViT) (Dosovitskiy et al., 2020) have shown impressive results on large-scale benchmarks, but they require substantially more data to outperform CNNs. ViT lacks the inductive bias of convolutions (translation equivariance, local connectivity), meaning it must learn spatial relationships from scratch. On a dataset of ten thousand images, a ViT would almost certainly overfit, whereas a CNN pre-trained on ImageNet arrives with that spatial knowledge built in. For completeness, the report will discuss ViT and its limitations in this context, but it will not be included in the experimental pipeline.

6 Ethical and Data Protection Considerations



Figure 8: The Staffordshire Bull Terrier — restricted under Irish law and frequently confused with the American Pit Bull Terrier and the American Staffordshire Terrier. This morphological ambiguity is the central ethical challenge: a misclassification can have life-or-death consequences for the animal.

6.1 The Stakes of Misclassification

Unlike most image classification projects, this one has direct consequences for living beings. A false positive (an unrestricted dog classified as restricted) could lead to seizure, mandatory muzzling, or legal proceedings against the owner. A false negative (a restricted dog classified as unrestricted) could leave the public at risk. Neither error is acceptable, but they carry different costs. The methodology explicitly addresses this through threshold optimisation: adjusting the decision boundary to reflect the relative severity of each type of error.

6.2 Data Sourcing and Compliance

All training data is sourced from the Stanford Dogs Dataset (Khosla et al., 2011), which is derived from ImageNet and released for academic research. No personally identifiable information is present in the dataset — the images depict dogs, not people. No web scraping of social media, personal websites, or enforcement databases is conducted.

The project complies with GDPR principles (Voigt and von dem Bussche, 2017):

- **Purpose limitation:** data is used solely for breed classification research.
- **Data minimisation:** only images necessary for the classification task are retained.
- **Storage limitation:** all data is stored in the researcher’s Google Drive and will be deleted upon project completion.

6.3 Bias Awareness

The Stanford Dogs Dataset has known limitations. It was compiled from ImageNet, which over-represents certain breeds, angles, and photographic styles. Dogs are typically shown in well-lit, centred compositions — conditions rarely matched in enforcement scenarios (poor lighting, motion blur, partial occlusion). The model’s performance on curated test images will therefore represent an upper bound; real-world deployment would require additional validation on field photographs.

Breed representation is also uneven. Some restricted breeds have 150+ images in the dataset; others (e.g., Staffordshire Bull Terrier) may have fewer. Data augmentation partially compensates, but cannot create diversity that does not exist in the source.

6.4 Transparency and Interpretability

The project adopts a transparency-by-design approach:

- All code is versioned and will be published alongside the report.
- Every model decision can be interrogated via Grad-CAM.
- The report will include a “failure gallery” showing the model’s worst misclassifications, not just its best performances.
- Confidence scores accompany every prediction — the model never says “this is a Rottweiler” without saying “and I am 87% sure.”

6.5 Deployment Safeguards

This project is a proof of concept, not a production system. However, the report will discuss the governance required for deployment: human-in-the-loop review, appeal processes, regular retraining, and the principle that an algorithm should assist, never replace, a qualified assessor.

7 Expected Challenges and Mitigation Strategies



(a) German Shepherd (restricted)



(b) Bull Mastiff (restricted)

Figure 9: Two restricted breeds with very different morphology. The model must learn that “restricted” is a legal category, not a visual one — these dogs share a legal status but not a body plan.

7.1 Challenge 1: Visual Similarity Between Classes

Some restricted and unrestricted breeds are nearly identical in appearance. A Staffordshire Bull Terrier (restricted) and a Boxer (unrestricted) share similar body proportions, coat types, and facial structure. The model must learn subtle discriminative features — or acknowledge uncertainty when the visual signal is ambiguous.

Mitigation: Grad-CAM analysis will reveal whether the model has learned meaningful features or is relying on spurious correlations. Threshold optimisation will allow the model to express uncertainty rather than forcing a binary decision.

7.2 Challenge 2: Heterogeneous Positive Class

The restricted class is defined by law, not biology. A German Shepherd and a Japanese Akita are both restricted but share few visual features. This makes the positive class inherently heterogeneous, complicating the learning task.

Mitigation: Transfer learning from ImageNet provides rich, general-purpose features that can represent diverse breeds within a single class. The model learns “restricted-ness” as a combination of breed-specific patterns, not a single visual template.

7.3 Challenge 3: Limited Dataset Size

With approximately ten thousand images total and around two thousand in the restricted class, the dataset is small by deep learning standards. Training from scratch would overfit within a few epochs.

Mitigation: Transfer learning reduces the number of parameters that need to be learned from scratch. Data augmentation artificially expands the effective training set. Early stopping prevents overfitting. The scaling experiment (50%, 75%, 100% of data) will quantify exactly how sensitive the model is to dataset size.

7.4 Challenge 4: Computational Resources

Training six to ten models, some with hyperparameter search across 50–100 trials, requires significant GPU time. A single training run of InceptionResNetV2 at 299×299 takes approximately 20–30 minutes on a Colab T4 GPU.

Mitigation: Google Colab Pro provides access to T4 and A100 GPUs with extended session times. All intermediate results are saved to Google Drive, allowing training to resume after session timeouts. The Hyperband algorithm is specifically designed for efficient search, pruning unpromising configurations early.

7.5 Challenge 5: Missing Breeds

Several restricted breeds — American Pit Bull Terrier, Japanese Tosa, Bandog, and XL Bully — are absent from the Stanford Dogs Dataset. The model cannot learn to recognise breeds it has never seen.

Mitigation: The report will clearly document which breeds are included and which are absent, with an honest discussion of the implications for real-world deployment. Future work could incorporate web-scraped images of under-represented breeds, subject to appropriate licensing and quality control.

8 Conclusion



Figure 10: The Bandog — a restricted breed that is not a single standardised breed but a type of large, muscular cross, historically bred as a guard dog. The absence of a standardised phenotype makes visual classification particularly challenging, even for human experts.

This proposal outlines a controlled investigation into the feasibility of automated restricted-breed classification using deep learning. The project is not a search for the highest accuracy number — it is a systematic comparison of architectures, regularisation strategies, and evaluation metrics, grounded in a real legal problem that affects real animals and their owners.

The design reflects three guiding principles:

Reproducibility. Every random seed is fixed. Every dataset split is deterministic. Every hyperparameter search trial is logged. Another researcher with access to the Stanford Dogs Dataset and a Colab account should be able to reproduce every result in this project.

Interpretability. No model is deployed as a black box. Grad-CAM visualisations accompany every prediction. The report includes both the model’s best performances and its worst failures. Confidence scores are reported alongside binary predictions.

Honesty. The project acknowledges what it cannot do. It cannot classify breeds absent from the training data. It cannot guarantee performance on field photographs taken in poor conditions. It cannot resolve the deeper question of whether breed-specific legislation is just. What it can do is provide evidence — rigorous, transparent, reproducible evidence — about what current technology can and cannot achieve in this domain.

The project contributes to three audiences simultaneously. For the machine learning community, it provides a benchmark comparison of transfer learning architectures on a novel binary classification task. For policy makers and enforcement agencies, it demonstrates the potential and limitations of automated breed identification. For the general public, it offers a readable account of how deep learning works, what it can see, and where it struggles — told through the lens of a

question that matters to anyone who has ever walked past a dog and wondered what breed it was.

8.1 Timeline

Table 4: Projected project timeline.

Phase	Duration
Data acquisition, cleaning, and splitting	1 week
Architecture timeline experiments (AlexNet to ResNet)	1 week
Modern transfer learning zoo (6 models)	1–2 weeks
Hyperparameter search (Keras Tuner)	1 week
Evaluation, threshold optimisation, Grad-CAM	1 week
Report writing and revision	2–3 weeks

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